Experiment 10.2

Analysis of Skip Distance and Tropospheric Propagation using MATLAB

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Aim

To analyse the skip distance of sky wave propagation for different angles of incidence and compare it with tropospheric line-of-sight propagation.

Software Used

MATLAB

Theory / Formulas

Skip distance increases with the angle of incidence for a given ionospheric layer height.
Tropospheric LOS distance depends on the antenna heights.
Coverage ratio = Skip distance / LOS distance.

Procedure

1. Open MATLAB and create a new script.
2. Define the parameters and generate the signals/data.
3. Compute the skip distance for angles of incidence from 10 to 70 degrees.
4. Compute the tropospheric LOS distance and the coverage ratio.
5. Plot and compare the sky wave and tropospheric propagation.

Output

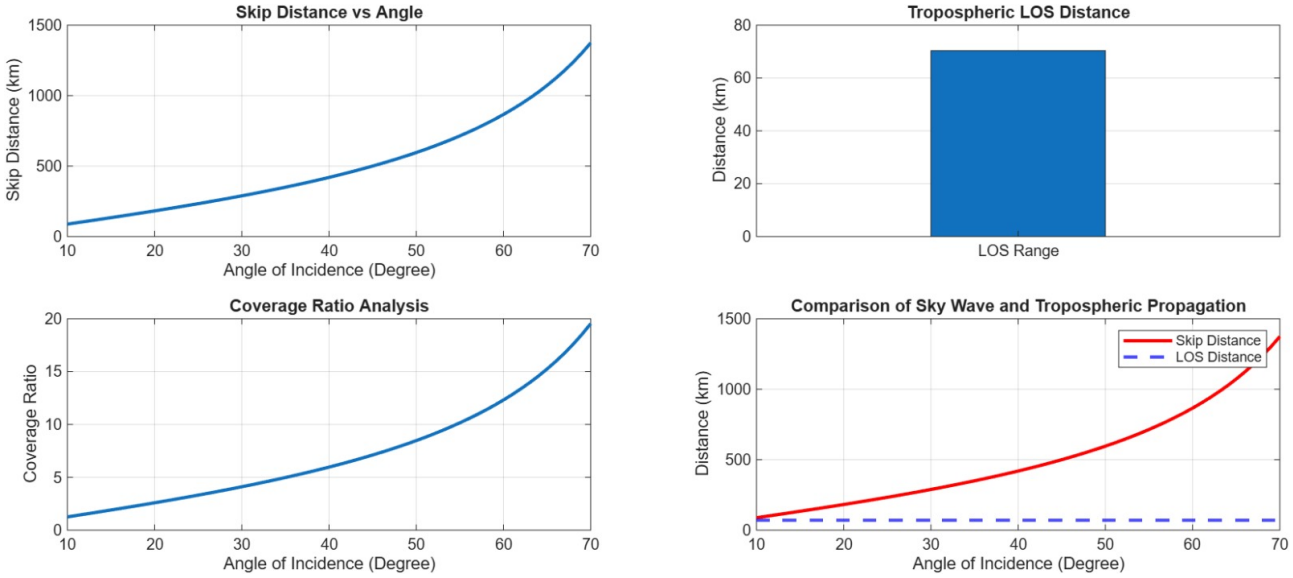


Fig. 1: Skip distance vs angle, tropospheric LOS distance, coverage ratio and comparison of sky wave and tropospheric propagation

Result

The skip distance rises with the angle of incidence and is much larger than the tropospheric LOS distance, giving a coverage ratio that grows with the angle.